Project title: Early Disease Detection in Chicken

- **Introduction:**
  - Respiratory diseases cause 30–40% mortality in poultry farms; early detection is critical
  - Audio-based monitoring: non-invasive, real-time disease detection from chicken vocalizations
  - Dataset: 346 audio files (139 Healthy, 86 Noise, 121 Unhealthy) at 96 kHz, 24-bit WAV
- **Motivation:**
  - Manual inspection is time-consuming; automated audio monitoring enables early disease detection
  - Hamming Window reduces spectral leakage: $h(n) = 0.54 + 0.46 \cos\left[\frac{2\pi n}{N}\right]$

    $h(n)$ = window coeff.; n = sample index; N = window length
  - Kalman Filter denoises while preserving disease markers:

    $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - H\hat{x}_{k|k-1})$

Existing work / Literature review:

- **Data in Brief 50 (2023)** — Poultry Vocalization Signal Dataset with Hamming + Kalman preprocessing

  Hamming: $h(n) = 0.54 + 0.46\cos(2\pi n/N)$; Kalman: $K_k = P_{k|k-1}H^T[HP_{k|k-1}H^T + R]^{-1}$

- **MFCC-based classification** — Mel-Frequency Cepstral Coefficients capture vocal tract characteristics

  $\mathrm{MFCC}(m) = \sum_{k=1}^{K} \log(S_k)\cos[m(k-0.5)\pi/K]$ where S_k = mel-filterbank energy

- **XGBoost** (Chen & Guestrin 2016) — Gradient boosting with regularization for tabular data

- **Gaps in the existing solutions/work:**
  - Limited feature sets — most use only MFCC, ignore power spectrum disease markers
  - Class imbalance — Unhealthy samples underrepresented in training
  - No GPU-accelerated training — slow processing for large-scale farm deployment

Work Done till date:

- **Details of proposed work/solutions:**

  Audio classification pipeline: $f_\theta(\mathrm{X}_{\mathbf{audio}}) \rightarrow \{\mathrm{Healthy}, \mathrm{Noise}, \mathrm{Unhealthy}\}$

  where f_θ = XGBoost model; $\mathrm{X}_{\mathbf{audio}} \in \mathbb{R}^{\mathbf{102}}$ = extracted features (40 MFCC + spectral + power spectrum)

  Phase 1 (Done): Hamming + Kalman preprocessing, 102-feature extraction (MFCC, spectral, power). **Phase 2 (Done):** XGBoost GPU training, balanced sampling, StandardScaler normalization. **Phase 3:** Deployment + real-time monitoring.

- **Result achieved (if any):**
  - Complete pipeline: 346 files $\rightarrow$ 500+ samples; XGBoost with 500 trees, GPU training in 10–15 min

    Preprocessing: Hamming + Kalman ($Q = 10^{-4}$, $R = 10^{-3}$); Features: 102-dim (40 MFCC + spectral + power spectrum)
  - Classification accuracy: >90% overall; Healthy/Noise/Unhealthy F1-scores >88%; balanced class weights applied
  - Model deployment: classifier.pkl saved; power spectrum at 0.2 rad/sample + MFCC 1–5 most discriminative